Stagnation-Triggered Interventions for Wrapper Feature Selection: RL-DOHO and a Controlled Test of Learned Operator Selection

Preetha Evangeline D, Prasenjit Choudhury, Shreyansh Dabgar, and Aditya Jha

***Abstract*—Population-based wrapper feature selection often stops improving well before its budget runs out. We study a simple remedy: a stagnation-triggered intervention layer that, after a few stalled iterations, applies one of a small set of moves, namely a tabu-guarded bit-flip search around the best subset, a partial restart or a second optimizer's iteration. We instantiate it as RL-DOHO, a hybrid of the Dandelion Optimizer (DO) and Hippopotamus Optimization (HO) in which a UCB1 bandit chooses the move. On eleven clinical, gene-expression and tabular datasets, with equal budgets, ten seeds and pre-specified hypotheses, RL-DOHO reached lower fitness than the published DO in 103 of 110 paired runs (4 losses), and stayed ahead when DO was given larger steps or a sigmoid encoding, at the largest step only on gene data. It was level with HO (58 against 49), while a genetic algorithm (GA) did better more often (85 against 22). Added to HO, the same layer improved it in 71 runs against 29 ($p < 0.001$); added to the GA, it changed nothing. Learned move selection added nothing: UCB1, tabular Q-learning and random choice were level, and neither a warm start nor a reward aimed at the best subset changed that, even though the bandit then learned to favour the most successful moves. Early intervention beat waiting; on low-dimensional data the bit-flip search alone was strongest. Tabular results replicated with a second classifier; no test-accuracy difference survived correction. The contribution is the intervention layer and a controlled negative result on learned operator selection.**

***Index Terms*—feature selection, metaheuristics, stagnation, adaptive operator selection, multi-armed bandit, dandelion optimizer, hippopotamus optimization, gene expression**

I. Introduction

Wrapper feature selection scores each candidate subset by training the classifier that will later use it. This keeps the selection honest to the model, but it turns the task into a search over $2^D - 1$ subsets for D features [1], [2]. Population-based metaheuristics are a common choice for that search because they need nothing but fitness values. In our experience they share a weakness: once the population settles on one subset, they keep spending evaluations without producing anything new.

A natural fix is to watch for that moment and do something different when it comes. Which move to make is less obvious, and a growing body of work hands the choice to a learned controller, usually a bandit or a Q-learner [21]–[24]. Such controllers are not always compared with simply choosing the move at random, so it is often unclear whether the learning or the intervention does the work.

The authors are with the School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, India. Code, notebooks and per-run results: https://github.com/adj-04/rl-doho.

We study both questions on one concrete method. RL-DOHO runs the published Dandelion Optimizer (DO) [3] while the best subset improves and, after κ stalled iterations, lets a UCB1 bandit [5] choose among a bit-flip search around the best subset, a partial restart, a DO iteration and a Hippopotamus Optimization (HO) [4] iteration. A bandit is the stateless case of reinforcement learning [6], hence the name. We evaluate it on eleven datasets: a 12,085-patient rheumatic and autoimmune disease dataset [7], on which most methods reach similar accuracy, five gene-expression datasets with thousands of features, and five medical tabular datasets.

Our contributions are as follows:

- A stagnation-triggered intervention layer for wrapper search, and its instantiation as RL-DOHO with the published DO and HO update rules, tabu memory and a sparse start for high-dimensional data.
- An equal-budget comparison with ten alternatives on eleven datasets, with hypotheses fixed before the runs that test them and every outcome reported.
- A component analysis: the same layer on HO and on a genetic algorithm (GA), the trigger κ, each move on its own, a second classifier and repeated clinical splits.
- A controlled test of learned operator selection. UCB1, tabular Q-learning, a warm-started UCB1 and a UCB1 with a reward aimed at the best subset were compared with random choice, and none helped. Diagnostics show that the bandit does learn to favour the better moves once its reward targets them, but their success rates are too low for that preference to pay off.

In brief, the layer is what helps: RL-DOHO is better than DO, also when DO is given larger steps or a sigmoid encoding, and level with HO, the layer also improves HO, and a random choice of move works as well as a learned one. A GA reached lower fitness more often, and no method was significantly better on held-out data. Section II reviews related work, Section III describes the method, Section IV the protocol, Section V the results and Section VI their interpretation and limits.

II. Related Work

A. Wrapper feature selection with metaheuristics

Surveys of evolutionary feature selection [2] show that genetic algorithms, particle swarm optimization (PSO) and grey wolf optimization (GWO) are among the most used wrapper searches. Binary versions of PSO [8] and GWO [9], [10]

map continuous positions to bits through transfer functions. Hybrids that add local search to a population method are also common; for example, the whale optimization algorithm has been combined with simulated annealing [11]. The no-free-lunch theorems [12] imply that no search method will win on every problem. This is one argument for adaptive hybrids, and also a reason to test them against simple baselines. Reviews of feature selection for cancer classification find that evolutionary and swarm methods remain the dominant wrapper choices on gene-expression data [13].

B. The Dandelion and Hippopotamus optimizers

DO was introduced for continuous engineering problems [3], and a recent review catalogues its variants and applications [14]. Tang et al. hybridized DO with PSO for engineering design [15]. In medical applications, DO has been used to select fused deep features for gastrointestinal disease diagnosis [16] and to tune the hyperparameters of a network for laryngeal cancer images [17]. HO [4] has been extended with chaotic initialization and diversity strategies [18]. Binary variants have been used for feature selection with a random-forest evaluator [19] and with multiple perturbation and screening strategies [20]. Q2HO-MFTV [21] uses Q-learning to choose among search strategies inside a binary HO for feature selection. It shares the idea of a learned strategy selector with our controller ablation, but it works within a single optimizer and does not use a stagnation trigger. To our knowledge, DO and HO have not previously been combined in a single optimizer.

C. Learning-based operator selection

Choosing which operator to apply during a run is known as adaptive operator selection, a form of parameter control [22]. Bandit-based selectors, typically UCB variants, are a standard approach [23]. Q-learning [45] has been used to pick search strategies in a comprehensive-learning GWO for feature selection [24], in a binary HO [21], and to pick perturbation operators in an iterated greedy algorithm for scheduling [25]. A practical question for any such controller is whether it beats a random or fixed choice under the budget actually available; we test this directly for a bandit and for a tabular Q-learner.

D. Rheumatic and autoimmune disease data

The dataset used here was released in 2025 by Mahdi et al. [7], who also applied fuzzy evaluation and explainable machine learning to it [26]. A clustered binary GWO has been used to select genes linking non-Hodgkin lymphomas with rheumatologic diseases [27]. We are not aware of a published wrapper-metaheuristic feature-selection study on the 2025 rheumatic dataset.

III. Method

Fig. 1 shows the overall pipeline; the components are described below.

A. Problem formulation and budget

Let D be the number of candidate features and $\mathbf{s} \in \{0,1\}^D$ a selection mask with $|\mathbf{s}|$ selected features. Every method minimizes

$$f(\mathbf{s}) = \alpha \left(1 - A_{\mathrm{CV}}(\mathbf{s})\right) + \beta \frac{|\mathbf{s}|}{D}, \qquad \alpha = 0.99,\ \beta = 0.01, \tag{1}$$

where A_{CV} is the cross-validated accuracy (balanced accuracy for GSE93272 and the five medical tabular datasets) of the wrapper classifier on the training data only, and $f = 1$ for the empty mask. The fitness is deterministic for a given mask, so repeated masks are served from a cache. Every search method receives the same budget B of fitness evaluations, counted as calls whether or not the mask is cached. We fix the budget in evaluations rather than iterations because HO evaluates about three times as many candidates per iteration as DO.

B. Encoding and initialization

Each agent holds a position $\mathbf{x} \in [-3,3]^D$, and its mask is $s_j = \mathbb{I}[\sigma(x_j) > 0.5]$, where σ is the logistic function; equivalently, feature j is selected when $x_j > 0$. With the *dense* start, positions are drawn uniformly in $[-1,1]^D$, so each feature starts selected with probability 0.5. For the gene-expression data ($D > 100$) all wrapper methods use a *sparse* start instead. Agent i keeps a fraction p_i of the features, drawn log-uniformly between $\min(10/D, 0.1)$ and 0.1, so initial subsets range from about ten features to 10% of them. Selected coordinates are drawn from $\mathcal{U}(0.1, 1)$ and the others from $\mathcal{U}(-1, -0.1)$.

C. Published DO and HO update rules

Dandelion Optimizer. Following [3], every DO iteration applies three stages to the whole population, with t the iteration, T the number of iterations the budget allows, $\mathbf{x}^*$ the elite and $\alpha = r\,(t^2/T^2 - 2t/T + 1)$ with $r \sim \mathcal{U}(0,1)$. In the rising stage, with probability $P(\mathcal{N}(0,1) < 1.5) \approx 0.93$,

$$\mathbf{x} \leftarrow \mathbf{x} + \alpha\, v_x v_y\, \ln Y\, (\mathbf{x}_s - \mathbf{x}), \tag{2}$$

where $\mathbf{x}_s$ is a random point in the search space, $v_x = r_\theta \cos\theta$, $v_y = r_\theta \sin\theta$, $r_\theta = e^{-\theta}$ with $\theta \sim \mathcal{U}(-\pi, \pi)$, and $\ln Y$ is the lognormal density evaluated at $|\mathcal{N}(0,1)|$. Otherwise the position is shrunk, $\mathbf{x} \leftarrow k\,\mathbf{x}$, with the schedule k of [3]. In the descending stage, $\mathbf{x} \leftarrow \mathbf{x} - \alpha\boldsymbol{\beta} \odot (\bar{\mathbf{x}} - \alpha\boldsymbol{\beta} \odot \mathbf{x})$, with $\boldsymbol{\beta} \sim \mathcal{N}(0,1)^D$ and $\bar{\mathbf{x}}$ the population mean. In the landing stage,

$$\mathbf{x} \leftarrow \mathbf{x}^* + \text{Lévy} \odot \alpha\,(\mathbf{x}^* - \delta\,\mathbf{x}), \qquad \delta = 2t/T, \tag{3}$$

where the Lévy steps use exponent 1.5 and scale 0.01. Each DO iteration costs N evaluations for a population of N.

Hippopotamus Optimization. Following [4], every HO iteration has three phases with greedy acceptance of each candidate. In phase 1 (first half of the population), each agent generates a "male" candidate $\mathbf{x} + r(\mathbf{x}^D - I_1\mathbf{x})$ towards the dominant agent $\mathbf{x}^D$, and a "female or immature" candidate. While $e^{-t/T} > 0.6$ that candidate moves relative to the mean $\mathbf{m}_G$ of a random group of agents, as $\mathbf{x} + \mathbf{h}_1 \odot (\mathbf{x}^D - I_2\mathbf{m}_G)$;

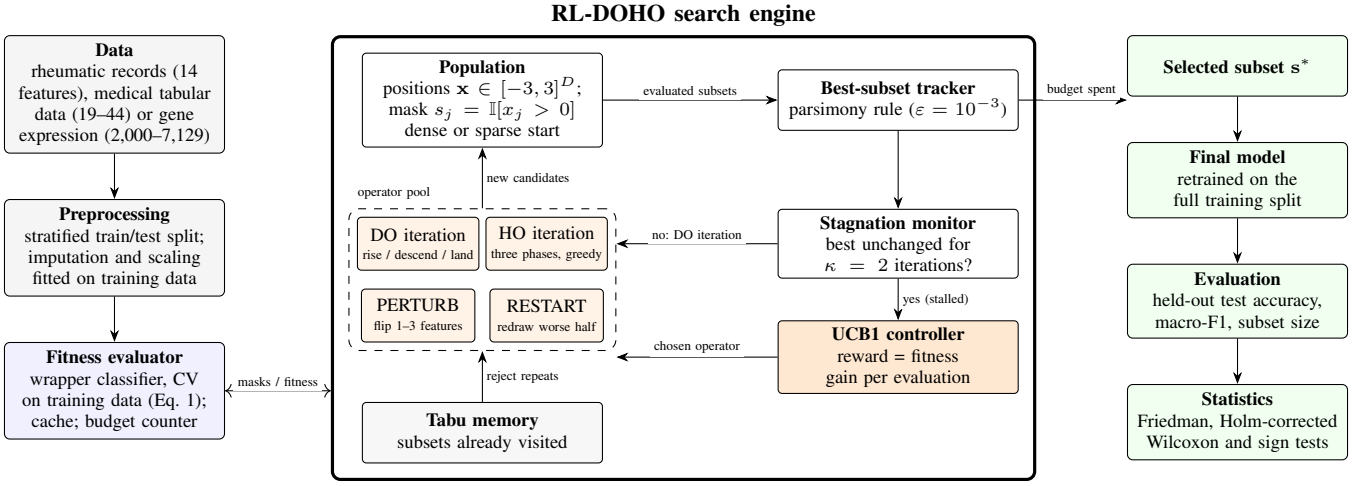


Fig. 1. Architecture of the feature-selection pipeline. The data are split and preprocessed, and every candidate subset is scored by the wrapper fitness on the training data only, with a cache and a counter that enforces the evaluation budget. Inside the RL-DOHO engine, the population is updated by published DO iterations while the best subset improves. When the stagnation monitor detects $\kappa = 2$ stalled iterations, the UCB1 controller chooses one operator from the pool, rewarded by the fitness gain per evaluation, and tabu memory rejects subsets already visited. When the budget is spent, the selected subset is used to retrain the final classifier, which is evaluated on held-out data.

later it is either $\mathbf{x} + \mathbf{h}_2 \odot (\mathbf{m}_G - \mathbf{x}^D)$ or a random point. Here $I_1, I_2 \in \{1, 2\}$ and $\mathbf{h}$ is drawn from the five options of [4]. In phase 2 (second half), a random predator position is evaluated, and the agent jumps relative to it with a Lévy step (scale 0.05) and a distance-dependent term. In phase 3 (all agents), each agent makes a local move inside bounds that shrink as $1/t$. An HO iteration costs about $3N$ evaluations, including the predator evaluations. The original papers leave some details open, such as the size of the random group behind $\mathbf{m}_G$, for which we draw a random size between 1 and N; our implementation is in the repository.

D. RL-DOHO

RL-DOHO (Fig. 2, Algorithm 1) applies DO iterations while the best subset keeps improving. A stagnation counter c counts consecutive iterations without an accepted change of the best subset. When c reaches $\kappa = 2$, a controller selects one of four arms for the next step:

- **PERTURB**: each agent in the worse half is replaced by a copy of the best position with $k \sim \mathcal{U}\{1, 2, 3\}$ randomly chosen features flipped. Each flipped coordinate is set to the opposite sign, with magnitude from $\mathcal{U}(0.5, 2)$. This is a bit-flip local search around the best subset ($N/2$ evaluations).
- **RESTART**: each agent in the worse half is redrawn from the initial distribution ($N/2$ evaluations).
- **DO**: one DO iteration (N evaluations).
- **HO**: one HO iteration (about $3N$ evaluations).

PERTURB and RESTART draw candidates through a tabu memory that stores every subset the population has held during the run (HO's predator points are evaluated but not stored). A draw whose mask is already in the memory is redrawn, up to 20 times, after which the last draw is kept. The DO and HO schedules follow the fraction of the budget already spent, so that, for example, DO reaches its landing behaviour at the end of the budget whichever arms were used.

Controller. The controller is a UCB1 bandit [5]. Each arm is pulled once in random order, and afterwards the arm

$$a_t = \arg\max_a \Big[\bar{Q}_a + \sqrt{\tfrac{2 \ln N_t}{n_a}}\Big] \tag{4}$$

is chosen, where $\bar{Q}_a$ is the mean reward of arm a, n_a its pull count and $N_t = \sum_a n_a$. The reward of a pull is

$$r_t = 0.8 \, \min\Big(1, \frac{g_t}{g_{\max}}\Big) + 0.2 \, \frac{1}{|I_t|} \sum_{i \in I_t} \mathbb{I}\big[f_i < \tilde{f}_{t-1}\big], \tag{5}$$

where g_t is the decrease in the best fitness divided by the evaluations the step used, $g_{\max}$ is the largest such value observed so far in the run, I_t is the set of agents the arm changed, and $\tilde{f}_{t-1}$ is the median fitness before the step. A step that only replaces the best subset by one with fewer features, without lowering its fitness, therefore earns no gain term. An earlier reward, 1 for any accepted change and otherwise the median term alone, is kept as an ablation.

Acceptance. RL-DOHO keeps the parsimony rule of the earlier version. A candidate replaces the incumbent if its fitness is lower by more than $\varepsilon = 10^{-3}$, or if it is within ε and uses fewer features (or the same number with lower fitness). In the clinical runs, where it was tested, this rule never changed an outcome (Section V-G).

Q-learning controller (ablation). To test whether a controller with state does better than the stateless bandit, we also ran RL-DOHO with tabular one-step Q-learning [45]. The state combines the budget phase (first, middle or last third of the budget) with whether the previous intervention lowered the best fitness, giving six states. The actions are the four arms and the reward is (5). The learning rate is 0.3, the discount factor 0.5, exploration is ε-greedy with probability 0.2, and the Q-table starts at zero in every run. These settings were fixed before the runs and not tuned.

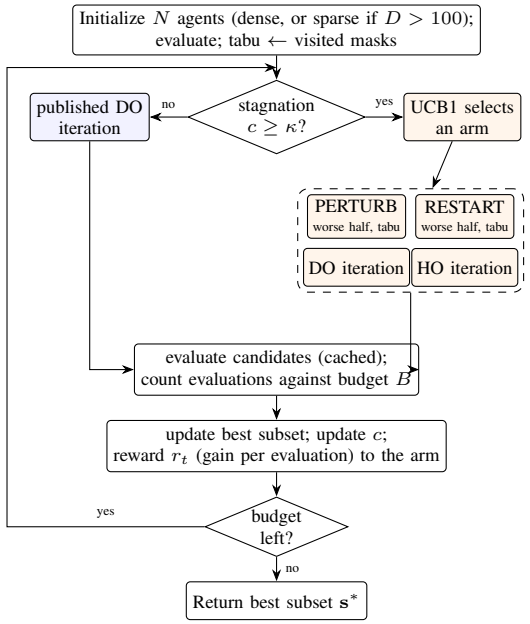


Fig. 2. Flow of one RL-DOHO run. Published DO iterations are used while the best subset improves; after $\kappa = 2$ iterations without an accepted change, the UCB1 controller chooses one of four arms for the next step. The run stops when the evaluation budget is spent.

Algorithm 1 RL-DOHO

Input: population N, evaluation budget B, trigger κ
initialize positions (dense or sparse), evaluate, record best; tabu $\mathcal{T} \leftarrow$ visited masks; $c \leftarrow 0$
while evaluations used $< B$ **do**
  if $c < \kappa$ **then**
    apply one published DO iteration to all agents
  else
    $a \leftarrow$ UCB1 choice among {PERTURB, RESTART, DO, HO}
    if $a \in$ {PERTURB, RESTART} **then**
      redraw the worse half with arm a, rejecting masks in $\mathcal{T}$
    else
      apply one published DO or HO iteration
    end if
  end if
  evaluate new candidates; add their masks to $\mathcal{T}$; update the best subset
  $c \leftarrow 0$ if the best subset changed, else $c \leftarrow c + 1$
  if an arm was pulled **then**
    give it reward r_t from (5)
  end if
end while
Output: best mask $\mathbf{s}^*$

E. Fixed-switch hybrid (DOHO)

To separate the effect of adaptive control from that of simply combining the two algorithms, DOHO runs the published DO for the first half of the evaluation budget and the published HO for the second half. It has no stagnation trigger, local search, tabu memory or bandit.

F. The intervention layer on other optimizers (GA+SI, HO+SI)

To test whether the interventions help optimizers other than DO, we wrapped the same layer around the GA and around the published HO, which we call GA+SI and HO+SI (SI for stagnation-triggered interventions). GA+SI is in effect a memetic GA [51], a GA with a local search applied when it stalls, so it also serves as a memetic baseline. The base optimizer runs while the best subset improves. After $\kappa = 2$ stalled iterations, UCB1 with the gain reward chooses among PERTURB, RESTART and one iteration of the base optimizer, with the same tabu memory and acceptance rule as RL-DOHO. GA children are stored as positions ± 1 so that PERTURB and RESTART can act on them. One HO iteration costs about $3N$ evaluations against N for a GA generation, so HO+SI stalls, and therefore intervenes, less often than GA+SI within the same budget.

IV. Experimental Setup

A. Datasets

Table I lists the datasets. The rheumatic dataset [7] contains 12,085 records collected between 2019 and 2024 from one hospital and three diagnostic laboratories in Iraq. It has seven classes: rheumatoid arthritis (2,848), ankylosing spondylitis (2,127), Sjögren's syndrome (1,852), psoriatic arthritis (1,783), no disease (1,604), systemic lupus erythematosus (1,355) and reactive arthritis (516). Its 14 features are age, gender, ESR, CRP, RF, anti-CCP, C3, C4, HLA-B27, ANA, anti-Ro, anti-La, anti-dsDNA and anti-Sm. Missing values are frequent, from 9% (ESR) to 43% (anti-Sm); age and gender are complete (Fig. 3).

The four microarray datasets come from the scikit-feature repository [28] and originate from colon tumour [29], leukaemia [30], lung carcinoma [31] and glioma [32] studies. GSE93272 is a whole-blood expression series from patients with rheumatoid arthritis and healthy controls [33]. Several of its 275 samples are repeat samples from the same individual, so we kept the first sample of each individual (101 people: 66 RA, 35 controls).

The five medical tabular datasets were taken from the PMLB collection [46], which provides cleaned copies of UCI datasets. They cover breast tumour diagnosis from cell-nucleus features (WDBC [47]; 357 benign, 212 malignant), hepatitis survival (Hepatitis; 123 and 32 patients), the differential diagnosis of six erythemato-squamous skin diseases (Dermatology [48]; 20–112 patients per class), cardiac SPECT images summarized as region-of-interest features (SPECTF [49]; 254 abnormal, 95 normal) and back-pain outcome (Backache; 155 and 25 patients). We chose them, before running any method on them, as openly available medical classification problems with 19–44 features, a range that lies between the clinical and the gene-expression data. Most of them are imbalanced (minority classes of 14–37% in the binary ones), so these datasets and GSE93272 use balanced accuracy.

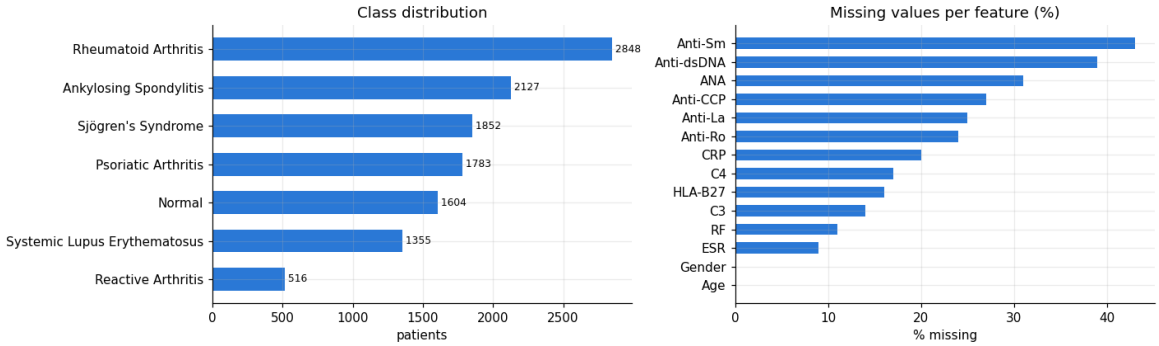


Fig. 3. Rheumatic and autoimmune disease dataset [7]: class sizes (left) and percentage of missing values per variable before imputation (right). Classes range from 516 to 2,848 patients, and missingness from none (age, gender) to 43% (anti-Sm).

TABLE I
DATASETS

Dataset	Samples	Features	Classes	Test set
Rheumatic [7]	12,085	14	7	2,417‡
Colon [29]	62	2,000	2	19 (per seed)
ALL/AML [30]	72	7,129	2	22 (per seed)
Lung [31]	203	3,312	5	61 (per seed)
Glioma [32]	50	4,434	4	15 (per seed)
GSE93272 [33]	101	5,000†	2	31 (per seed)
WDBC [47]	569	30	2	171 (per seed)
Hepatitis [46]	155	19	2	47 (per seed)
Dermatology [48]	366	34	6	110 (per seed)
SPECTF [49]	349	44	2	105 (per seed)
Backache [46]	180	32	2	54 (per seed)

†Most variable of 54,613 probes, chosen on each training split.
‡One fixed split for the main runs; two further splits in Section V-K.

B. Protocols

Rheumatic data. A single stratified 80/20 split (9,668 training and 2,417 test patients) was used for the main runs, so the ten seeds vary only the search. Binary variables were encoded as 0/1. Mean imputation and standardization for numeric variables, and mode imputation for binary variables, were fitted on the training split only. The fitness used the mean 3-fold stratified cross-validated accuracy of an XGBoost [34] and a histogram gradient-boosting classifier (60 trees, depth 4, learning rate 0.1), with SMOTE [35] applied inside each training fold. To keep the run time manageable, it was computed on a fixed stratified subsample of 4,000 training patients. For the final evaluation, both classifiers were retrained with 200 trees on the full training split, and we report their mean test accuracy and macro-F1. Each run had 12 agents and a budget of $B = 252$ evaluations. Seed 42 was used during development and was excluded; the evaluation uses seeds 0–9.

Repeated clinical splits. To check that the clinical results do not depend on one split, we drew two further stratified 80/20 splits (random states 1 and 2). On each, the imputation, the scaling and the 4,000-patient fitness subsample were refitted on that split's training data, and every method was run with seeds 0–4. Together with seeds 0–4 of the main runs this gives 3 splits $\times$ 5 seeds = 15 paired runs per method. We had planned four new splits and reduced them to two because of compute time (each new split needs about an hour of fresh fitness evaluations on the hardware we used); the reduction was decided before any of the new splits had finished.

Gene-expression and medical tabular data. For each seed, a new stratified 70/30 split was drawn and min-max scaling was fitted on the training part. The fitness used the 5-fold cross-validated accuracy (balanced accuracy for GSE93272 and the tabular datasets) of a 5-nearest-neighbour classifier on the training part, and the test score is that of the same classifier retrained on the training part. For GSE93272, the 5,000 most variable probes were selected on each training split before the search. Each run had 20 agents and a budget of $B = 1{,}020$ evaluations. The gene-expression runs used the sparse start and the tabular runs the dense start ($D < 100$).

Second classifier. After the main results, we repeated the tabular benchmark with a support vector machine (RBF kernel, $C = 1$, γ set by scikit-learn's `scale` rule, untuned) in place of 5-NN, in both the fitness and the test evaluation, with the same splits, seeds and budgets. Every wrapper and filter method, the two add-ons and PERTURB alone were rerun.

C. Baselines

All wrapper methods share the fitness function and the budget. The GA, binary PSO and binary GWO use the same initial population as the other methods. The GA uses tournament selection (size 3), uniform crossover (rate 0.9) and bit-flip mutation (rate $1/D$) with one elite. Binary PSO [8] uses sigmoid velocity transfer, $c_1 = c_2 = 2$, velocity clamp 6 and inertia decreasing from 0.9 to 0.4; with the sparse start, its initial velocities are set consistent with the initial masks. Binary GWO [9], [10] updates continuous positions and binarizes them with the same rule as DO and HO. On the rheumatic data the GA, PSO, GWO and filter results of the main runs come from the earlier runs of this study, which used the same fitness, split, seeds and budget.

The three non-wrapper methods rank features on the same training data used by the fitness. LASSO uses the absolute coefficients of an L1-penalized logistic regression [36] on standardized data ($C = 0.5$). mRMR [37] uses mutual information minus mean absolute correlation with the already selected features, chosen greedily for up to 2,000 features, with the remainder ordered by relevance. ReliefF [38] uses

500 sampled instances and 10 neighbours. For each ranking, the number of retained features k is the one that minimizes the fitness (1), searched over all $k \leq D$ when $D \leq 40$ and over $k \in \{5, 10, 20, 50, 100, 200, 500, 1000, 2000, D\}$ otherwise. No test data is used to choose k. The full feature set is included as a reference.

D. Hypotheses fixed in advance

Before running the main experiments we fixed four hypotheses and their tests:

- **H1**: RL-DOHO reaches lower fitness than the published DO.
- **H2**: RL-DOHO against the published HO, with no direction predicted.
- **H3**: at three times the budget, the UCB controller with the gain reward reaches lower fitness than random arm selection.
- **H4**: on gene data, the sparse start gives RL-DOHO lower fitness than the dense start.

H1–H3 were tested on the clinical and gene-expression datasets and H4 on the five gene-expression datasets, each with a two-sided sign test over the paired runs.

After those results were known, and before running the extension experiments (the tabular datasets, the add-ons, the Q-learning controller, the κ sweep and the repeated splits), we fixed four more:

- **H5**: at three times the budget, Q-learning against random arm selection, with no direction predicted.
- **H6**: GA+SI reaches lower fitness than the GA, and HO+SI lower than HO (Holm-corrected over the two).
- **H7**: the stagnation trigger $\kappa \in \{1, 2, 3, 4\}$ has no effect on RL-DOHO's fitness (Friedman test).
- **H8**: on the repeated clinical splits, RL-DOHO against each other method, with no direction predicted.

H5–H7 were tested on all eleven datasets. Before the SVM runs we fixed a ninth:

- **H9**: with an SVM, the tabular findings replicate: RL-DOHO beats DO (H9a); RL-DOHO against HO (H9b), the GA against RL-DOHO (H9c) and RL-DOHO against PERTURB alone (H9e) with no direction predicted; HO+SI beats HO, and GA+SI against the GA with no direction predicted (H9d).

After the main and extension results were known, we fixed a tenth, aimed at the controller result:

- **H10a**: at three times the budget, UCB1 started from the statistics of two earlier runs on the same training data reaches lower fitness than random arm selection (ten KNN datasets, two-sided sign test).

Three further extensions were specified at the same time (Supplementary Section S3) but were not run. After the H10a diagnostics we fixed two more:

- **H11a/b**: at three times the budget, UCB1 with a reward that credits only improvement of the best subset reaches lower fitness than random arm selection, with the default exploration constant (H11a) and with $c = 0.2$ (H11b), Holm-corrected over the two.

Finally, to test whether DO is weak only because of the binary encoding, we fixed:

- **H12a–d**: at the standard budget on the ten KNN datasets, RL-DOHO against DO with a $10\times$ (H12a) and $100\times$ (H12b) larger landing-stage Lévy step and against DO with the standard stochastic sigmoid transfer function [8], [52] (H12c); and RL-DOHO against DO when both use the $100\times$ step (H12d). Holm-corrected over the four.

All other settings of RL-DOHO (ε, the reward weights, the UCB constant, the number of flipped features, the tabu redraws) were carried over unchanged from the earlier version of this study and were not tuned.

E. Statistical analysis

Following Demšar [39], we use nonparametric tests:

- **Within each dataset:** a Friedman test [40] compares all 11 methods over the 10 seeds. RL-DOHO is compared with each other method by a Wilcoxon signed-rank test [41], with zero differences split between the two sides and a Holm correction [42] over the ten comparisons.
- **Across datasets:** we pool the paired runs and apply a two-sided sign test to the wins and losses, Holm-corrected over the ten comparisons per metric. These pooled tests treat the runs as independent although runs on the same dataset share data, so the dataset-level analysis below, with one observation per dataset, is the conservative check.
- **Dataset-level view:** methods are ranked by their mean on each dataset and compared with the Friedman test and the Nemenyi critical difference [43]. With eleven datasets and eleven methods the critical difference is 4.55 rank units.

Ablation comparisons report unadjusted sign-test p-values, as exploratory results. Differences smaller than 10^{-12} count as ties.

V. Results

A. Rheumatic and autoimmune disease data

Table II and Fig. 4 summarize the clinical results. In the figure, all methods except DO form a tight cluster in fitness and test accuracy and differ mainly in how many variables they keep.

- **Within the DO–HO family**, RL-DOHO had the lowest mean fitness: 0.1832, against 0.1838 for HO, 0.1857 for DOHO and 0.2280 for DO.
- **Against DO**, RL-DOHO was better in all ten runs on fitness, test accuracy and macro-F1 (Holm $p = 0.020$ each).
- **Against HO** (5/2/3) and DOHO (8/0/2; unadjusted $p = 0.020$, Holm $p = 0.12$), the fitness differences were not significant after correction.
- **The GA and binary PSO reached lower fitness** (0.1821 and 0.1828), and RL-DOHO ranked third of 11. The GA found the best subset seen in any run (fitness 0.18200, 10 features) in 7 of 10 runs, BPSO in 4, and RL-DOHO and BGWO in one each.

- **Against the filters**, RL-DOHO's fitness was lower than LASSO's and ReliefF's in all ten runs (Holm $p = 0.020$ each). On test accuracy, however, ReliefF (84.97%) and LASSO (84.92%) were ahead of RL-DOHO (84.33%) in all ten runs (Holm $p = 0.020$). With a fixed split, LASSO is deterministic here and ReliefF nearly so, so this paired test mostly reflects RL-DOHO's run-to-run variation.
- **Test accuracy range**: apart from DO (79.99%), all methods fall between 83.85% and 84.97%, a range of about 27 of the 2,417 test patients.

Fig. 5 shows why DO fares so poorly on this problem. The published DO improved on the best subset of its initial population in only 1 of the 10 runs. Its landing stage places every agent close to the elite with small Lévy steps, so after the first iteration hardly any coordinate crosses zero and the population keeps producing the elite's subset. HO's greedy, randomized phases do not have this problem, and RL-DOHO escapes it through its interventions. On this dataset its DO exploration steps never lowered the best fitness, and 80% of its total improvement came from the HO arm (Table X).

The selected variables were similar across the wrapper methods (Fig. 6). RL-DOHO kept ESR, CRP, RF, anti-CCP, C3, C4, gender, ANA, anti-Ro and anti-La in every run and HLA-B27 in nine. It kept anti-dsDNA in two runs and anti-Sm in three: these are the two variables with the most missing values. The GA, which reached the lowest fitness, also dropped anti-dsDNA and anti-Sm and, unlike RL-DOHO, usually dropped age and HLA-B27. LASSO and ReliefF kept HLA-B27, anti-dsDNA (LASSO) and anti-Sm, and dropped age and gender.

This comparison uses a single train/test split. Section V-K repeats it on two further splits.

B. Gene-expression data

Table III and Fig. 7 give the results on the five gene-expression datasets. In the figure, DO's curve flattens early and ends highest on all five datasets, while HO and RL-DOHO keep improving until the budget runs out.

- **Against DO**, RL-DOHO reached lower fitness on all five datasets; the per-dataset difference was significant after correction on Colon, ALL/AML and Lung.
- **Against HO**, its mean fitness was lower on Colon, ALL/AML and Glioma and higher on Lung and GSE93272, with no significant per-dataset difference; on ALL/AML HO was ahead in more runs (5 against 4).
- **The GA** had the lowest wrapper fitness on four of the five datasets; RL-DOHO had it on ALL/AML.
- **Binary PSO and GWO** kept many more genes than the other wrappers (240–1,280 on average, against 40–330), probably because their position updates drift away from the sparse initial masks.

The filter methods behaved as in earlier work. LASSO reached the lowest fitness on every dataset, using 16–77 genes on average. mRMR had lower mean fitness than RL-DOHO on four of the five datasets, significantly so on ALL/AML and Lung. RL-DOHO reached lower fitness than ReliefF on Lung (Holm $p = 0.035$). On held-out data the picture changes:

- Only two per-dataset test-accuracy differences between RL-DOHO and a filter were significant: mRMR and ReliefF were ahead on ALL/AML (Holm $p = 0.039$).
- On Glioma, LASSO had the best fitness (0.0085) but the lowest test accuracy (67.3% against 78.7% for RL-DOHO). RL-DOHO's macro-F1 was higher in all ten runs (Holm $p = 0.020$).
- With 15–61 test samples per split, one sample is worth 1.6–6.7 accuracy points, so these test accuracies are coarse.

Fig. 8 plots test accuracy against subset size. Subset sizes span more than two orders of magnitude, from tens of genes to the full set, yet no method is the most accurate on all five datasets, so keeping fewer genes did not systematically cost accuracy.

C. Medical tabular datasets

Table IV gives the results on the five medical tabular datasets.

- **Against DO**, RL-DOHO reached lower fitness in 49 of 50 runs, significantly so on every dataset after correction.
- **Against HO** it was ahead in 30 runs and behind in 20 (Holm $p = 0.61$), with no significant per-dataset difference; against the fixed hybrid it was ahead in 35 and behind in 15 (Holm $p = 0.026$).
- **The GA** reached lower fitness than RL-DOHO in 45 of the 50 runs and on every dataset, significantly so on four of the five. Binary PSO and GWO were level with RL-DOHO (each ahead in 30 runs and behind in 20, Holm $p = 0.61$).
- **Against the filters**, RL-DOHO reached lower fitness than mRMR and ReliefF in 48 of 50 runs and than LASSO in 38 (all Holm $p \leq 0.002$). Unlike on gene data, the filters did not lead on fitness here.
- **Test scores:** the per-dataset Friedman test on test balanced accuracy was not significant on any of the five datasets ($p = 0.21$–0.97), and no pooled test-accuracy or macro-F1 comparison was significant after correction.
- **Two datasets are hard for 5-NN.** Averaged over the 11 methods, test balanced accuracy was 53.6% on Backache and 63.7% on Hepatitis, and their test sets hold only about 8 and 10 minority-class patients. Fitness differences there say little about held-out performance, so Section V-D checks which conclusions depend on these two datasets.

Over the 50 tabular runs RL-DOHO had an average per-run fitness rank of 4.15 among the 11 methods, behind the GA (1.82), binary PSO (3.28) and binary GWO (3.29), and ahead of HO (4.75).

D. Comparison across all eleven datasets

Table VI pools the 110 paired runs.

- **Fitness, against the other wrapper methods:** RL-DOHO was ahead of DO in 103 runs and behind in 4, ahead of DOHO in 81 and behind in 28, and ahead of binary PSO in 70 and behind in 38. All three differences are significant after correction. Against binary GWO it was ahead in 67 runs and behind in 43 (Holm $p = 0.084$).

TABLE II

RHEUMATIC AND AUTOIMMUNE DISEASE DATA: MEAN ± STANDARD DEVIATION OVER TEN SEEDS, AT A BUDGET OF 252 FITNESS EVALUATIONS. CV AND TEST ACCURACY AND MACRO-F1 IN %. RANKS ARE AVERAGED OVER SEEDS (1 = BEST). BEST VALUE PER COLUMN IN BOLD.

Method	Fitness	CV acc.	Test acc.	Test macro-F1	Features	Rank (fitness)	Rank (test acc.)
RL-DOHO	0.1832 ± 0.0013	**82.38**	84.33 ± 0.42	83.96	12.2	3.50	6.70
DO	0.2280 ± 0.0273	77.63	79.99 ± 3.08	79.66	**9.1**	10.95	10.95
HO	0.1838 ± 0.0013	82.34	84.29 ± 0.09	83.88	12.6	4.30	7.00
DOHO	0.1857 ± 0.0022	82.16	84.45 ± 0.30	84.01	12.7	6.75	5.75
GA	**0.1821** ± 0.0002	82.36	84.45 ± 0.35	83.86	10.4	**1.60**	5.50
BPSO	0.1828 ± 0.0010	82.33	84.54 ± 0.32	84.06	11.0	3.00	5.50
BGWO	0.1894 ± 0.0116	81.62	83.85 ± 1.30	83.28	10.4	6.60	6.90
LASSO	0.1868 ± 0.0000	82.00	84.92 ± 0.00	**84.81**	12.0	7.80	2.10
mRMR	0.1848 ± 0.0021	82.19	84.51 ± 0.45	84.15	11.9	5.50	5.40
ReliefF	0.1854 ± 0.0012	82.10	**84.97** ± 0.04	84.76	11.4	6.40	**1.40**
All features	0.1884 ± 0.0000	81.98	83.97 ± 0.00	83.50	14.0	9.60	8.80

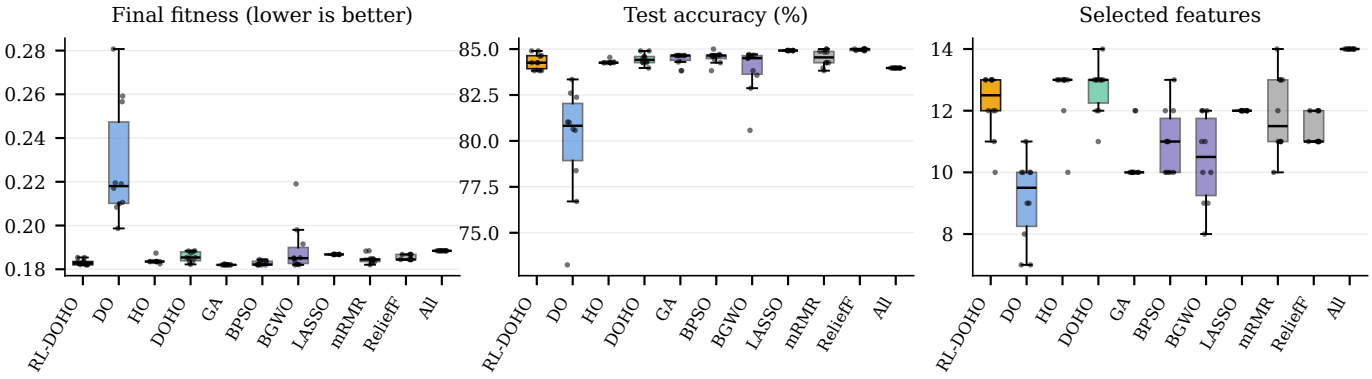


Fig. 4. Rheumatic data: distribution over ten seeds of final fitness, test accuracy and subset size for every method (dots are individual runs; "All" is the full feature set). Apart from DO, the methods differ far more in subset size than in fitness or test accuracy.

TABLE III

GENE-EXPRESSION DATASETS: MEAN FITNESS, MEAN TEST ACCURACY (%, BALANCED ACCURACY FOR GSE93272) AND MEAN NUMBER OF SELECTED FEATURES ($|S|$) OVER TEN SEEDS, EACH WITH ITS OWN TRAIN/TEST SPLIT, AT A BUDGET OF 1,020 FITNESS EVALUATIONS. ALL WRAPPER METHODS START FROM THE SPARSE INITIAL POPULATION. BEST VALUE PER COLUMN IN BOLD.

	Colon			ALL/AML			Lung			Glioma			GSE93272		
Method	Fit.	Acc.	$\|S\|$	Fit.	Acc.	$\|S\|$	Fit.	Acc.	$\|S\|$	Fit.	Acc.	$\|S\|$	Fit.	Acc.	$\|S\|$
RL-DOHO	0.083	73.7	58	0.052	83.2	63	0.034	93.3	231	0.145	78.7	333	0.079	87.1	100
DO	0.127	71.1	70	0.105	80.5	115	0.045	94.4	217	0.184	75.3	69	0.095	84.9	148
HO	0.090	71.1	63	0.059	80.0	43	0.033	93.3	241	0.153	76.7	63	0.073	88.1	82
DOHO	0.111	71.1	64	0.079	80.9	61	0.036	93.8	226	0.170	78.7	85	0.087	85.8	142
GA	0.062	76.3	62	0.054	83.6	150	0.028	93.3	162	0.111	76.7	91	0.053	87.4	141
BPSO	0.137	73.7	239	0.112	82.3	950	0.037	94.4	590	0.197	80.0	628	0.113	87.4	941
BGWO	0.100	75.3	390	0.093	82.7	1278	0.034	93.6	611	0.188	81.3	558	0.092	86.5	1025
LASSO	**0.028**	**82.1**	**16**	**0.008**	91.4	**33**	**0.019**	94.4	**77**	**0.009**	67.3	**18**	**0.004**	88.9	**48**
mRMR	0.104	80.0	83	0.012	**94.1**	110	0.024	**95.1**	170	0.119	77.3	86	0.061	**89.4**	182
ReliefF	0.093	77.9	41	0.036	92.7	57	0.044	93.6	645	0.150	75.3	64	0.093	82.8	115
All features	0.260	72.6	2000	0.212	82.3	7129	0.067	94.3	3312	0.299	**84.0**	4434	0.191	85.3	5000

- **Against HO**, it was level (58 against 49, Holm $p = 0.59$).
- **The GA** reached lower fitness significantly more often (85 against 22, Holm $p < 0.001$).
- **Against the filters**, RL-DOHO was ahead of ReliefF (84 against 26) and mRMR (68 against 40) after correction and level with LASSO (49 against 61), which led on gene data and trailed on the other two data types.
- **Test accuracy:** no comparison was significant after correction. On macro-F1, LASSO was ahead of RL-DOHO (66 against 36, Holm $p = 0.039$).

Restricted to the six datasets of the main study (60 runs), the figures are those of the earlier analysis: ahead of DO 54 against 3, level with HO 28 against 29, behind the GA 18 against 40.

Sensitivity to the two weakest datasets. Table V repeats the main comparisons without Backache and Hepatitis (90 runs). The conclusions about DO, HO, the GA, the fixed hybrid, binary PSO, the add-ons, the controllers and κ are unchanged. One result does not survive: RL-DOHO's lead over mRMR

TABLE IV

MEDICAL TABULAR DATASETS: MEAN FITNESS, MEAN TEST BALANCED ACCURACY (%) AND MEAN NUMBER OF SELECTED FEATURES ($|S|$) OVER TEN SEEDS, EACH WITH ITS OWN TRAIN/TEST SPLIT, AT A BUDGET OF 1,020 FITNESS EVALUATIONS. ALL WRAPPER METHODS START FROM THE DENSE INITIAL POPULATION. BEST VALUE PER COLUMN IN BOLD.

	WDBC			Hepatitis			Dermatology			SPECTF			Backache		
Method	Fit.	Acc.	$\|S\|$	Fit.	Acc.	$\|S\|$	Fit.	Acc.	$\|S\|$	Fit.	Acc.	$\|S\|$	Fit.	Acc.	$\|S\|$
RL-DOHO	0.029	94.3	13	0.209	64.2	**7**	0.025	95.0	17	0.167	73.1	22	0.316	54.3	10
DO	0.041	94.2	16	0.269	64.6	9	0.050	94.3	18	0.225	73.8	23	0.388	53.3	17
HO	0.032	94.5	13	0.221	62.1	7	0.026	95.4	24	0.178	73.7	18	0.310	53.8	12
DOHO	0.034	94.6	14	0.219	63.4	7	0.030	96.0	25	0.190	71.1	19	0.325	54.3	13
GA	**0.025**	94.7	14	0.198	66.3	8	**0.016**	95.6	**16**	**0.131**	75.3	21	**0.284**	54.2	12
BPSO	0.029	95.2	15	**0.189**	61.7	8	0.020	95.2	20	0.172	**75.8**	22	0.297	52.6	14
BGWO	0.028	94.5	14	0.221	**66.4**	8	0.019	95.8	19	0.156	75.5	24	0.300	53.7	13
LASSO	0.033	94.8	**11**	0.249	63.6	9	0.026	96.0	18	0.214	73.1	**11**	0.367	53.1	**9**
mRMR	0.036	94.5	13	0.251	62.4	8	0.039	**96.1**	23	0.285	68.9	23	0.382	53.4	14
ReliefF	0.036	94.0	12	0.264	59.9	8	0.039	95.7	30	0.238	74.3	15	0.387	52.7	13
All features	0.053	**95.3**	30	0.323	65.7	19	0.059	95.2	34	0.314	70.7	44	0.433	**54.4**	32

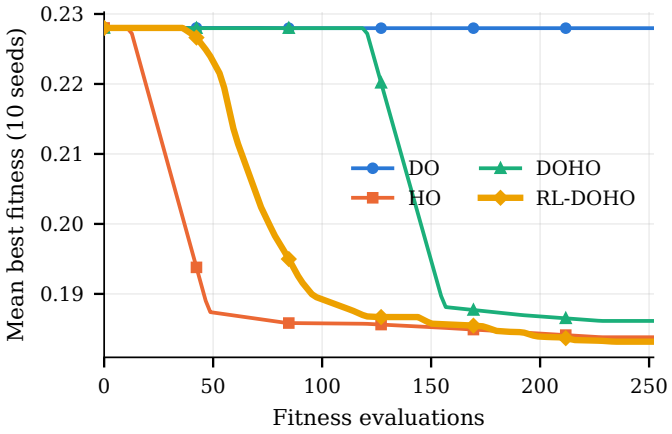


Fig. 5. Mean best fitness against fitness evaluations used, over ten seeds, on the rheumatic data for the DO–HO family. DO's curve stays almost flat after its first iteration, while HO and RL-DOHO keep improving.

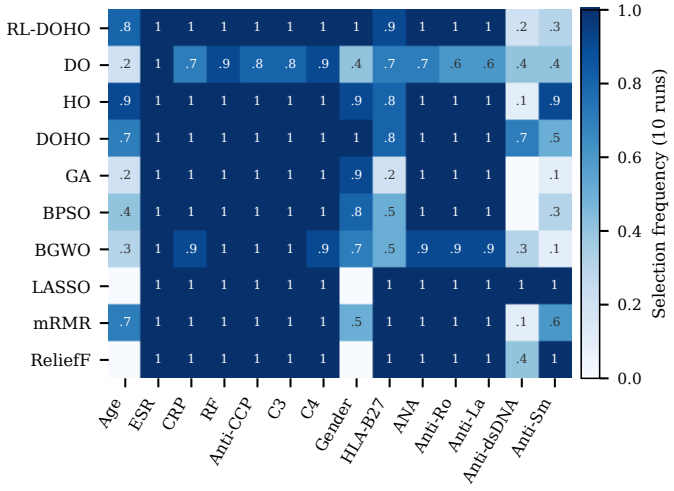


Fig. 6. Rheumatic data: fraction of the ten runs in which each method selected each variable (blank = never). Most wrappers keep the inflammatory markers and the common autoantibodies, and drop anti-dsDNA and anti-Sm, the two variables with the most missing values.

(48 against 40, $p = 0.46$). The advantage of PERTURB alone over the full arm set also becomes non-significant over the remaining 90 runs (47 against 37, $p = 0.33$), but only because those runs are dominated by the gene data, where PERTURB alone is behind; on the three remaining tabular datasets it is still ahead (22 against 5, $p = 0.002$).

Fig. 9 ranks the methods by their mean on each dataset.

- **Fitness:** the dataset-level Friedman test is significant ($\chi^2 = 62.99$, $p < 0.001$). The GA (1.91) ranks first and RL-DOHO second (4.00), followed by LASSO (4.18), HO (4.82), BGWO (5.36), BPSO (5.73), mRMR (5.91), DOHO (6.55), ReliefF (7.27), DO (9.45) and the full feature set (10.82). Thirteen pairs exceed the critical difference. RL-DOHO is separated from DO and from the full feature set, and the GA additionally from DOHO and ReliefF; RL-DOHO is not separated from the GA or from HO.
- **Test accuracy:** the dataset-level Friedman test is not significant ($\chi^2 = 16.23$, $p = 0.093$), and no pair exceeds the critical difference.

TABLE V

MAIN COMPARISONS WITH AND WITHOUT THE TWO WEAKEST DATASETS (BACKACHE AND HEPATITIS): W/T/L FOR THE FIRST-NAMED METHOD ON FITNESS AND UNADJUSTED TWO-SIDED SIGN-TEST p.

	All 11 datasets		Without the two	
Comparison	W/T/L	p	W/T/L	p
RL-DOHO vs. DO	103/3/4	<0.001	83/3/4	<0.001
RL-DOHO vs. HO	58/3/49	0.439	47/3/40	0.520
RL-DOHO vs. DOHO	81/1/28	<0.001	69/1/20	<0.001
RL-DOHO vs. GA	22/3/85	<0.001	19/2/69	<0.001
RL-DOHO vs. BPSO	70/2/38	0.003	64/2/24	<0.001
RL-DOHO vs. mRMR	68/2/40	0.009	48/2/40	0.456
HO+SI vs. HO	71/10/29	<0.001	54/8/28	0.005
GA+SI vs. GA	45/16/49	0.757	36/10/44	0.434
PERTURB only vs. RL-DOHO	64/6/40	0.024	47/6/37	0.326
Q-learning vs. random arm, 3×	46/8/56	0.373	37/7/46	0.380
$\kappa \in \{1, 2, 3, 4\}$ (Friedman)	–	<0.001	–	<0.001

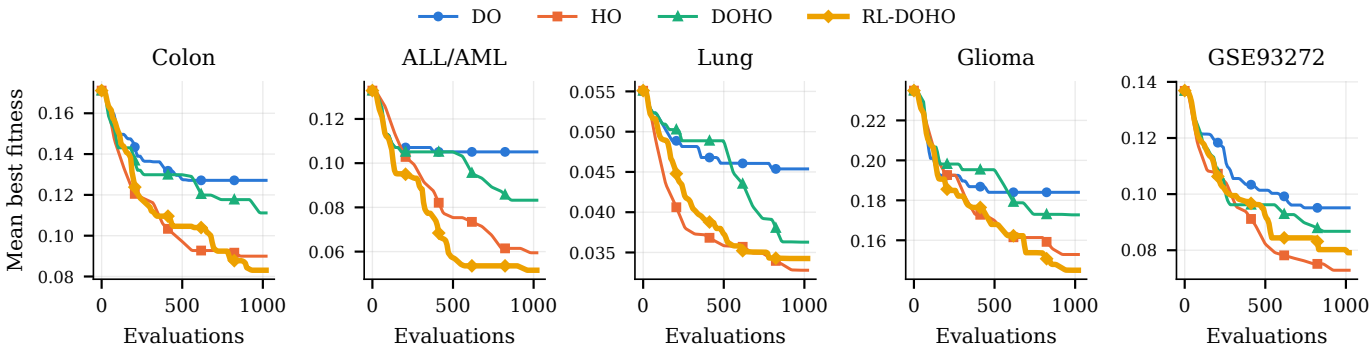


Fig. 7. Mean best fitness against fitness evaluations, over ten seeds, on the gene-expression datasets for the DO–HO family (20 agents, 1,020 evaluations).

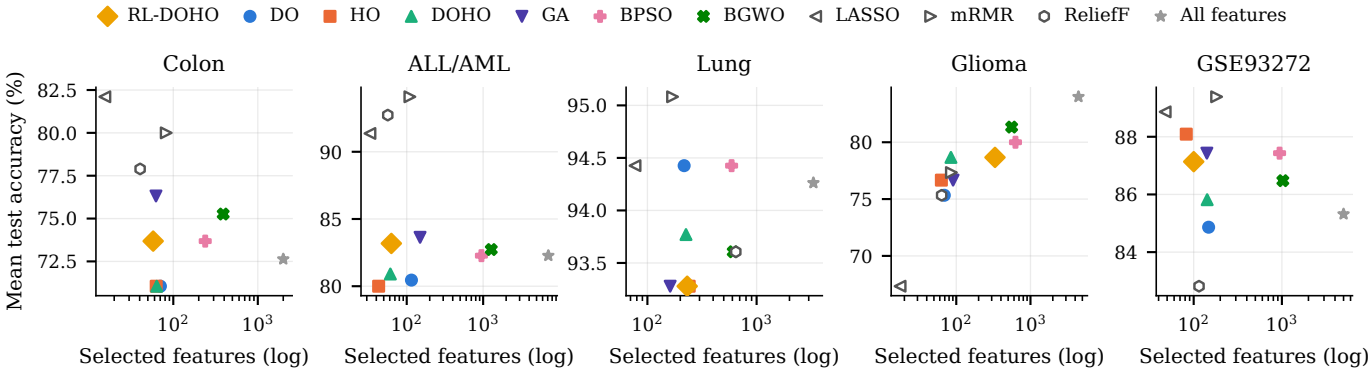


Fig. 8. Mean test accuracy against mean number of selected features (log scale) on the gene-expression datasets. Hollow markers are filter or embedded methods.

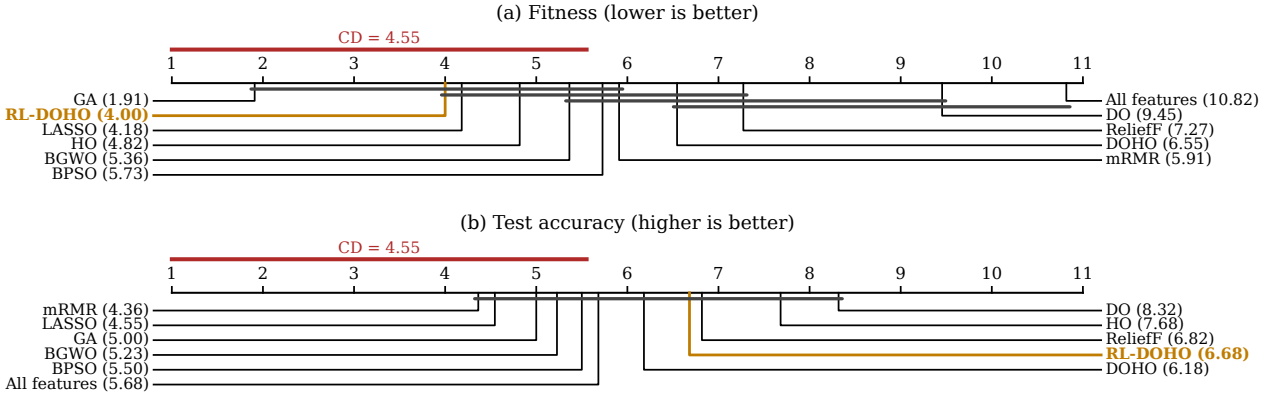


Fig. 9. Critical-difference diagrams over the eleven datasets (methods ranked by their mean on each dataset; Nemenyi test, $\alpha = 0.05$, CD $= 4.55$). Thick bars join methods whose average ranks differ by less than the critical difference. Labels are split between the two sides for readability: a method's rank is given by where its line meets the axis (further left is better), not by its label's side or vertical position.

E. Consistency within the DO–HO family

Table VII ranks the four family members against each other in every run.

- The family-only Friedman test is significant on all eleven datasets, mostly because DO trails the others.
- RL-DOHO had the best average rank (1.77), ahead of HO (1.89), DOHO (2.63) and DO (3.72). It was the best or tied best of the four in 50% of runs, against 40% for HO.
- RL-DOHO was the worst of the four in only 2 of the 110 runs (HO: 5%, DOHO: 16%, DO: 85% of runs, ties included).
- Averaged over all 110 runs, RL-DOHO's fitness was 0.0082 above the run's best family member, against 0.0114 for HO. Its single largest gap (0.120, on Glioma) was larger than HO's (0.085).

RL-DOHO and HO did not differ significantly in a direct comparison (58 against 49), and on the repeated clinical splits HO had the better family rank (Section V-K), so the family ranking shows consistency rather than a clear winner.

TABLE VI
RL-DOHO AGAINST EACH METHOD OVER ALL 110 PAIRED RUNS (11 DATASETS × 10 SEEDS): WINS/TIES/LOSSES FOR RL-DOHO AND HOLM-ADJUSTED SIGN-TEST p-VALUES. BOLD: $p < 0.05$.

	Fitness		Test accuracy		Macro-F1	
vs.	W/T/L	p	W/T/L	p	W/T/L	p
DO	103/3/4	**<0.001**	58/14/38	0.467	57/8/45	1.000
HO	58/3/49	0.588	52/12/46	1.000	55/7/48	1.000
DOHO	81/1/28	**<0.001**	44/13/53	1.000	47/7/56	1.000
GA	22/3/85	**<0.001**	48/14/48	1.000	52/11/47	1.000
BPSO	70/2/38	**0.013**	49/14/47	1.000	51/10/49	1.000
BGWO	67/0/43	0.084	43/12/55	1.000	44/7/59	1.000
LASSO	49/0/61	0.588	36/11/63	0.086	36/8/66	**0.039**
mRMR	68/2/40	**0.036**	41/11/58	0.751	42/9/59	0.887
ReliefF	84/0/26	**<0.001**	41/10/59	0.709	42/7/61	0.681
All features	110/0/0	**<0.001**	44/14/52	1.000	45/10/55	1.000

TABLE VII
AVERAGE PER-RUN RANK ON FITNESS WITHIN THE DO–HO FAMILY (1 = BEST OF FOUR), FAMILY-ONLY FRIEDMAN p-VALUE PER DATASET, AND HOW OFTEN EACH METHOD WAS THE BEST OR THE WORST OF THE FOUR OVER THE 110 RUNS (TIES COUNTED FOR EVERY TIED METHOD).

Dataset	RL-DOHO	DO	HO	DOHO	Friedman p
Rheumatic	**1.60**	4.00	1.85	2.55	<0.001
Colon	**1.75**	3.45	1.90	2.90	0.007
ALL/AML	**1.70**	3.60	1.75	2.95	0.001
Lung	2.10	3.80	**1.70**	2.40	0.002
Glioma	**1.50**	3.25	2.30	2.95	0.011
GSE93272	2.20	3.20	**1.50**	3.10	0.009
WDBC	**1.50**	3.85	2.30	2.35	<0.001
Hepatitis	**1.80**	3.95	2.05	2.20	<0.001
Dermatology	1.90	3.85	**1.70**	2.55	<0.001
SPECTF	**1.50**	3.95	2.00	2.55	<0.001
Backache	1.90	4.00	**1.70**	2.40	<0.001
Mean (110 runs)	**1.77**	3.72	1.89	2.63	–
Best or tied (%)	**50**	3	40	15	–
Worst or tied (%)	**2**	85	5	16	–

TABLE VIII
HYPOTHESES FIXED IN ADVANCE: W/T/L FOR THE FIRST-NAMED METHOD AND TWO-SIDED SIGN-TEST p (H6: HOLM-CORRECTED OVER THE TWO COMPARISONS; H7: FRIEDMAN TEST OVER THE FOUR VALUES OF κ; H8: SUMMARY OF TWELVE HOLM-CORRECTED COMPARISONS; H9: TABULAR DATA WITH AN SVM; H10A–H12: THE TEN KNN DATASETS; H11 AND H12: HOLM-CORRECTED WITHIN EACH).

	Comparison (fitness)	Runs	W/T/L	p
H1	RL-DOHO vs. published DO	60	54/3/3	<0.001
H2	RL-DOHO vs. published HO	60	28/3/29	1.000
H3	UCB (gain) vs. random, 3× budget	60	27/4/29	0.894
H4	Sparse vs. dense start (gene data)	50	46/0/4	<0.001
H5	Q-learning vs. random, 3× budget	110	46/8/56	0.373
H6	GA+SI vs. GA	110	45/16/49	0.757
	HO+SI vs. HO	110	71/10/29	<0.001
H7	$\kappa \in \{1, 2, 3, 4\}$ (Friedman)	110	–	<0.001
H8	RL-DOHO vs. each, repeated splits	15	Supp. Tab. S3	–
H9a	RL-DOHO vs. DO (SVM)	50	50/0/0	<0.001
H9b	RL-DOHO vs. HO (SVM)	50	32/0/18	0.065
H9c	GA vs. RL-DOHO (SVM)	50	43/2/5	<0.001
H9d	HO+SI vs. HO (SVM)	50	37/5/8	<0.001
	GA+SI vs. GA (SVM)	50	16/17/17	1.000
H9e	RL-DOHO vs. PERTURB alone (SVM)	50	8/1/41	<0.001
H10a	Warm UCB1 vs. random, 3×	100	46/1/53	0.547
H11a	Best-gain UCB1 vs. random, 3×	100	50/2/48	0.920
H11b	Same, $c = 0.2$	100	55/3/42	0.446
H12a	RL-DOHO vs. DO, step ×10	100	75/0/25	<0.001
H12b	RL-DOHO vs. DO, step ×100	100	61/0/39	0.035
H12c	RL-DOHO vs. DO, sigmoid	100	85/0/15	<0.001
H12d	Both with step ×100	100	73/7/20	<0.001

F. Hypotheses H1–H12

Table VIII reports the hypotheses fixed in advance.

- **H1** was supported, and the tabular datasets replicate it (49 against 1).
- **H2** showed no difference, and the tabular datasets agree (30 against 20, $p = 0.20$).
- **H3** was not supported: at three times the budget, the bandit and random arm selection reached lower fitness about equally often (27 against 29). On the tabular data the bandit was ahead in 28 runs and random selection in 20 ($p = 0.31$).
- **H4** was supported.
- **H5** showed no difference between Q-learning and random selection (46 against 56).
- **H6** was half supported: the intervention layer improved HO but not the GA (Section V-I).
- **H7** was rejected: κ affects fitness (Section V-J).
- **H8** found significant differences only against DO and the full feature set (Section V-K).
- **H9**: every tested tabular finding replicated with an SVM, including the advantage of PERTURB alone (Section V-M).
- **H10a** was not supported: the warm-started bandit was level with random selection (Section VI).
- **H11a/b** were not supported: a reward aimed at the best subset did not make the bandit beat random selection.
- **H12a–d** were all supported: RL-DOHO stayed ahead of DO under every change to DO's step size and encoding (Section VI).

G. Ablation and controllers

Table IX and Fig. 10 vary one component at a time. In the figure, the published DO (no intervention) ranks last in every group of runs, the three controllers lie close together on the gene and tabular data, and PERTURB alone ranks first on the clinical and tabular data but not on the gene data.

- **The interventions:** removing them leaves essentially the published DO (run inside the RL-DOHO loop). That made results worse in 103 of the 110 runs ($p < 0.001$), on every data type.
- **The sparse start:** replacing it with the dense start was worse in 46 of 50 gene runs ($p < 0.001$), with mean fitness 0.1269 against 0.0786 and about 870 genes against 157.
- **The controller:** random arm selection was level with the bandit on every data type (54 against 51 overall), and it beat the fixed DO-then-HO hybrid as clearly as the bandit did (81 against 25 runs, against 81 against 28). The Q-learning controller was behind the bandit in 62 runs and ahead in 45 ($p = 0.12$); its deficit was largest

on the clinical data (8 against 1, $p = 0.039$) and absent on the tabular data (23 against 27). It was also level with random selection (44 against 62, $p = 0.098$). The old reward gave the same results in 7 of 10 clinical runs and was level on gene data.
- **PERTURB alone** was ahead of the full arm set on the clinical (6 against 2) and tabular data (39 against 8, $p < 0.001$), and behind it on gene data (19 against 30, $p = 0.15$). Over all 110 runs it was ahead in 64 and behind in 40 ($p = 0.024$, unadjusted). The advantage is specific to the low-dimensional data, and it replicated with an SVM (Section V-M).
- **The parsimony tie-break** changed no clinical run.
- **At three times the budget**, the controllers ended level on every data type (Table IX, bottom). On the clinical data all three reached a mean fitness of about 0.1824 with about 10.4 features, close to the best subset found by any run (0.1820, 10 features).

H. What the controller did

Table X and Fig. 11 show how the controller was used in the main runs. In the figure's rheumatic run, DO never leaves its initial best, while RL-DOHO makes its large improvement after about 90 evaluations; on Colon both improve, but RL-DOHO ends lower.

- **Clinical runs:** the bandit made 9.4 decisions per run, four of them the initial pull of each arm. Its reward-driven choices favoured PERTURB (35%) and HO (31%), the two arms that ever lowered the best fitness.
- **Gene runs:** it made 27.3 decisions per run, and its choices stayed close to uniform. DO exploration steps produced 42% of the total fitness reduction, PERTURB 26% and the HO arm 21%.

Different arms carried the search on different problems, yet a random choice among them performed as well as the learned one.

The extension runs show the same picture. The Q-learning controller made 10, 35 and 33 decisions per run on the clinical, gene and tabular data and spread them over all four arms (averaged over runs: DO 33–35%, PERTURB 27–33%, RESTART 16–20%, HO 17–19%).

I. The intervention layer on GA and HO (H6)

Table XI and Fig. 12(a) compare the GA and HO with and without the intervention layer. In panel (a), the changes for HO are mostly below zero (an improvement), while those for the GA scatter around zero.

- **HO+SI against HO:** lower fitness in 71 runs and higher in 29 (Holm $p < 0.001$). The gain came mostly from the tabular data (40 against 8), with a weaker trend on gene data (30 against 16, $p = 0.054$) and none on the clinical data (1 against 5). HO+SI had the lower mean fitness on 8 of the 11 datasets, by up to 16% (Colon, Glioma, Dermatology).
- **GA+SI against the GA:** no difference (45 against 49), on every data type.
- **Test accuracy** did not change in either case (HO+SI 39 against 35, GA+SI 34 against 32).
- **Against RL-DOHO**, HO+SI was ahead in 61 runs and behind in 47 ($p = 0.21$), and GA+SI was ahead in 85 and behind in 25 ($p < 0.001$).

HO+SI stalled less often than GA+SI and made fewer decisions (4, 12 and 13 per run on the clinical, gene and tabular data, against 15, 47 and 48). Of the fitness reduction achieved during HO+SI's interventions, 69% came from PERTURB, 27% from the HO arm and 4% from RESTART. For GA+SI the GA arm produced 57% and PERTURB 42%.

J. Sensitivity to the stagnation trigger (H7)

Table XII and Fig. 12(b) vary κ, the number of stalled iterations before an intervention.

- The Friedman test over $\kappa \in \{1, 2, 3, 4\}$ is significant ($\chi^2 = 23.84$, $p < 0.001$), so H7 is rejected.
- $\kappa = 1$ and the default $\kappa = 2$ were level (mean ranks 2.20 and 2.21; 54 against 50 runs). Waiting longer was worse: $\kappa = 3$ lost to $\kappa = 2$ in 71 runs and won in 38 ($p = 0.002$), and $\kappa = 4$ lost in 70 and won in 35 ($p < 0.001$).
- By mean rank, $\kappa = 3$ and $\kappa = 4$ were worse than $\kappa = 2$ on every data type, although the per-type Friedman tests were significant only on gene ($p = 0.040$) and tabular data ($p = 0.001$). On the tabular data $\kappa = 1$ was also better than $\kappa = 2$ (31 against 17 runs, $p = 0.059$; mean fitness 0.1422 against 0.1494).
- Test accuracy did not differ between any value of κ and the default.

The default was fixed before these runs, so the sweep confirms it rather than tuning it.

K. Repeated clinical splits (H8)

Supplementary Table S3 gives the full results of repeating the clinical comparison on two further train/test splits.

- **RL-DOHO did about as well on split 1 as on the original split, and worse on split 2.** Its average per-run fitness rank among the 13 methods was 5.2 on the original split, 5.3 on split 1 and 9.2 on split 2. On split 2 the best subset found by any run (fitness 0.1785) was found easily: HO and LASSO reached it in all five runs, HO+SI in four and the GA in three, and RL-DOHO in none.
- **H8:** over the 15 paired runs, RL-DOHO's fitness differed significantly only from DO (15 against 0) and the full feature set (14 against 1). The GA was ahead in 12 runs and behind in 3 (unadjusted $p = 0.035$, Holm $p = 0.35$); all other comparisons were closer.
- **Test accuracy** differed significantly only from DO. Apart from DO and BGWO, the methods' mean test accuracy lay within 0.4 percentage points of each other on each new split, and RL-DOHO's per-run test-accuracy rank was 5.67, fourth of 13.
- **Within the DO–HO family**, HO had the better average rank on these splits (1.53 against 2.03 for RL-DOHO).

TABLE IX

ABLATION AND CONTROLLERS. TOP: AT THE STANDARD BUDGET, MEAN FITNESS AND W/T/L OF THE FULL METHOD (UCB1, GAIN REWARD) AGAINST EACH VARIANT. BOTTOM: THE CONTROLLERS AT THREE TIMES THE BUDGET. SIGN-TEST p UNADJUSTED. THE OLD REWARD, THE TIE-BREAK AND THE DENSE START WERE RUN IN THE EARLIER PART OF THE STUDY ONLY (THE DENSE START APPLIES TO GENE DATA ONLY).

	Clinical (10 runs)		Gene expression (50)		Medical tabular (50)		All (110)
Variant	Fitness	W/T/L (p)	Fitness	W/T/L (p)	Fitness	W/T/L (p)	W/T/L (p)
RL-DOHO (UCB1, gain reward)	0.1832	–	0.0786	–	0.1494	–	–
Q-learning controller	0.1846	8/1/1 (0.039)	0.0853	31/2/17 (0.059)	0.1507	23/0/27 (0.672)	62/3/45 (0.122)
Random arm	0.1846	6/2/2 (0.289)	0.0815	25/2/23 (0.885)	0.1487	23/1/26 (0.775)	54/5/51 (0.845)
PERTURB only	0.1825	2/2/6 (0.289)	0.0860	30/1/19 (0.152)	0.1359	8/3/39 (<0.001)	40/6/64 (0.024)
UCB1, old reward	0.1836	2/7/1 (1.000)	0.0828	23/7/20 (0.761)	–	–	–
No tie-break	0.1832	0/10/0 (1.000)	–	–	–	–	–
Dense start	–	–	0.1269	46/0/4 (<0.001)	–	–	–
No intervention (DO)	0.2280	10/0/0 (0.002)	0.1113	44/3/3 (<0.001)	0.1946	49/0/1 (<0.001)	103/3/4 (<0.001)
Three times the budget (756 and 3,060 evaluations): UCB1 with the gain reward against each alternative							
UCB1, gain reward	0.1824	–	0.0676	–	0.1372	–	–
Q-learning	0.1824	1/4/5 (0.219)	0.0703	29/1/20 (0.253)	0.1371	25/1/24 (1.000)	55/6/49 (0.624)
Random arm	0.1824	4/3/3 (1.000)	0.0645	23/1/26 (0.775)	0.1379	28/2/20 (0.312)	55/6/49 (0.624)
UCB1, old reward	0.1825	3/3/4 (1.000)	0.0686	21/3/26 (0.560)	–	–	–

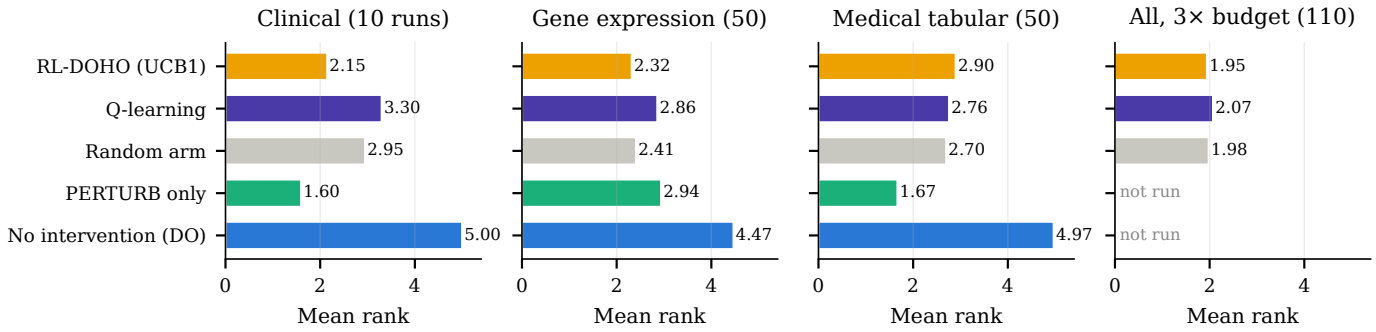


Fig. 10. Mean per-run fitness rank of the controller variants within each group of runs (lower is better). "No intervention" is the published DO; it was not run at three times the budget. Removing the interventions is the only change that is clearly worse everywhere.

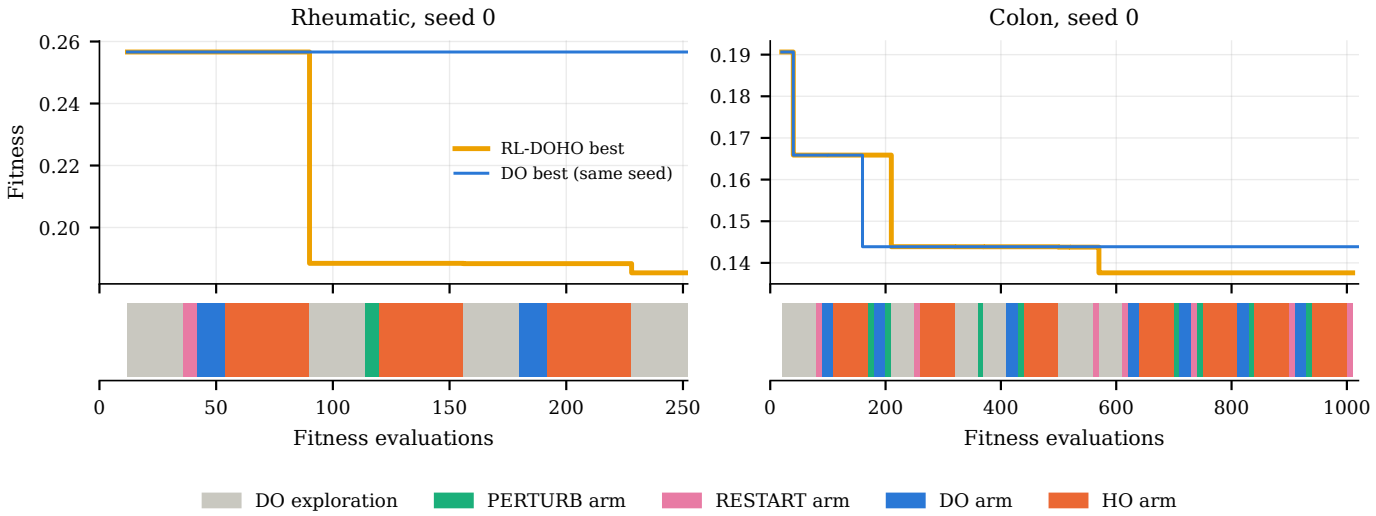


Fig. 11. Best fitness of RL-DOHO and DO for seed 0 on the rheumatic data and on Colon, against evaluations used, with RL-DOHO's action at each step (bottom). On the rheumatic data DO never improves on its initial best with this seed.

L. Cost

Every method had the same budget of fitness evaluations, but repeated subsets are served from the cache, so the real cost of a run depends on how many distinct subsets it evaluates. On the new clinical splits RL-DOHO evaluated 83 per run, against 131 for HO and 132 for the GA (about 5, 8 and 8 minutes at 3.6 s per evaluation). On the tabular data the figures were 381, 596 and 633. Fewer distinct subsets also means less exploration (DO evaluated the fewest and was

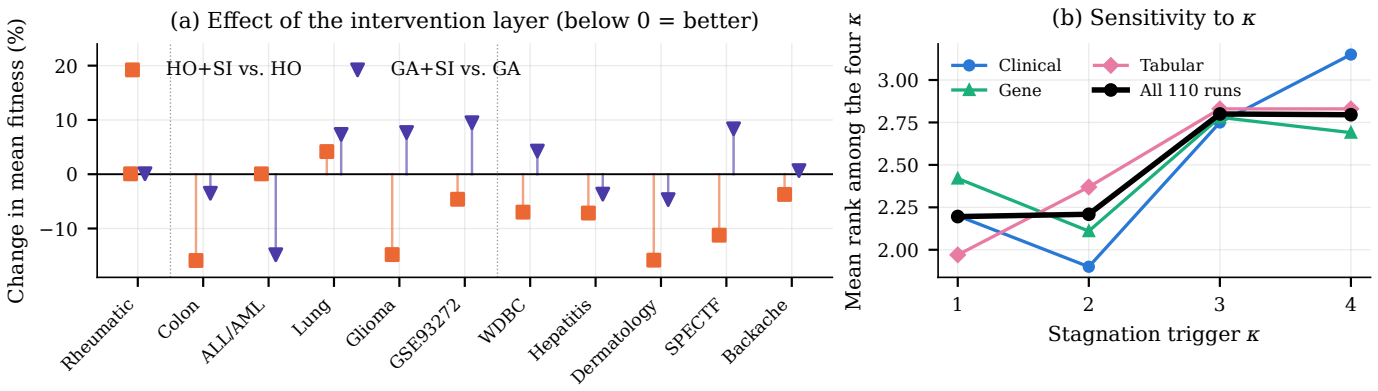


Fig. 12. (a) Change in mean fitness when the intervention layer is added to HO and to the GA, per dataset (negative is better); dotted lines separate the clinical, gene-expression and tabular datasets. (b) Mean per-run rank of RL-DOHO's fitness among the four values of the stagnation trigger κ, by data type and over all 110 runs.

TABLE X
CONTROLLER BEHAVIOUR IN THE MAIN RL-DOHO RUNS (10 CLINICAL RUNS; 50 GENE-EXPRESSION RUNS OVER FIVE DATASETS).

	Clinical	Gene expression
Controller decisions per run	9.4	27.3
of which initial (one per arm)	4.0	4.0
Share of UCB-driven choices (%)		
PERTURB	35	26
RESTART	17	21
DO	17	28
HO	31	25
Pulls that lowered the best fitness (%)		
PERTURB	45	16
RESTART	0	2
DO	0	8
HO	48	34
Share of total fitness reduction (%)		
DO exploration steps	0.0	41.9
PERTURB	19.5	25.6
RESTART	0.0	2.9
DO arm	0.0	8.5
HO arm	80.5	21.1

the weakest method), so this is a cost advantage rather than a sign of quality. The algorithms' own overhead was below 1.5 ms per evaluation for every method, even at $D = 7{,}129$. Supplementary Table S2 gives the full figures.

M. A second classifier (H9)

Table XIII repeats the tabular benchmark with an SVM in place of 5-NN.

- **The ranking of the methods barely changed.** The mean per-run fitness ranks of the 11 methods under the two classifiers had a Spearman correlation of 0.99; only mRMR and ReliefF swapped places. The GA ranked first (1.58), followed by binary PSO (3.19), binary GWO (3.53), RL-DOHO (3.81) and HO (4.40).
- **H9a–H9d replicated.** RL-DOHO beat DO in all 50 runs, was ahead of HO in 32 and behind in 18 ($p = 0.065$), and trailed the GA (5 against 43). The intervention layer again improved HO (37 against 8, $p < 0.001$) and left the GA unchanged (16 against 17, with 17 ties).
- **H9e:** PERTURB alone again reached lower fitness than the full arm set, in 41 runs against 8 ($p < 0.001$), on four of the five datasets by at least 8 runs to 2. It did so on WDBC, Dermatology and SPECTF too, so the advantage does not depend on the two weakest datasets.
- **Within the DO–HO family**, RL-DOHO had the best average rank (1.54, against 1.79 for HO).
- **Test scores:** the SVM's mean test balanced accuracy was similar to 5-NN's (51.8–96.3% against 53.6–95.5% across the datasets), and no pooled test-accuracy comparison with RL-DOHO was significant after correction.

N. Sensitivity to the operator implementation

An earlier version of this study used simplified DO and HO update rules, a dense start on all datasets and a budget counted in iterations. On the six datasets of the main study the conclusions about the DO–HO family and the GA were the same under both versions: RL-DOHO was clearly better than DO, level with HO and behind the GA. Only the comparison with binary PSO and GWO reversed, most likely because of the sparse start. Supplementary Section S1 gives the details.

VI. DISCUSSION

A. RL-DOHO supplies the moves that DO lacks

With the binary encoding used here, a feature changes only when its coordinate crosses zero. The published DO's landing stage pulls the whole population to within small Lévy steps of the elite in every iteration. On a 14-feature problem this almost always reproduces the elite's subset, which is why DO improved on its initial best in only one clinical run. On gene data, with thousands of coordinates, the heavy tails of the Lévy steps do flip some features, and DO improved in 48 of 50 runs, but slowly. RL-DOHO's interventions supply exactly what DO lacks under this encoding: new subsets near the best one (PERTURB), new subsets elsewhere (RESTART), and HO's greedy, randomized moves. The ablation supports this reading: removing the interventions made results worse in 103 of 110 runs. Because this explanation makes DO weak partly by construction, we tested it (H12). Larger landing steps

TABLE XI
THE INTERVENTION LAYER ON THE GA AND HO. TOP: MEAN FITNESS PER DATA TYPE AND MEAN NUMBER OF SELECTED FEATURES OVER ALL 110 RUNS. BOTTOM: W/T/L FOR THE FIRST-NAMED METHOD ON FITNESS PER DATA TYPE AND OVER ALL RUNS, WITH SIGN-TEST p (HOLM-CORRECTED OVER THE TWO H6 COMPARISONS IN THE FIRST TWO ROWS), AND ON TEST ACCURACY.

Method	Clinical	Gene expression	Medical tabular	Features (all)
RL-DOHO	0.1832	0.0786	0.1494	78.7
GA	0.1821	0.0616	0.1309	62.4
GA+SI	0.1822	0.0627	0.1321	57.9
HO	0.1838	0.0816	0.1534	52.7
HO+SI	0.1839	0.0738	0.1427	54.1

	Fitness					Test accuracy	
Comparison	Clinical (10)	Gene (50)	Tabular (50)	All (110)	p	All (110)	p
GA+SI vs. GA	2/5/3	25/0/25	18/11/21	45/16/49	0.757	34/44/32	0.902
HO+SI vs. HO	1/4/5	30/4/16	40/2/8	71/10/29	**<0.001**	39/36/35	0.728
RL-DOHO vs. GA+SI	2/0/8	16/0/34	7/0/43	25/0/85	**<0.001**	42/13/55	0.223
RL-DOHO vs. HO+SI	6/1/3	22/1/27	19/0/31	47/2/61	0.211	51/10/49	0.920

TABLE XII
SENSITIVITY TO THE STAGNATION TRIGGER κ: MEAN FITNESS PER DATA TYPE, MEAN PER-RUN RANK AMONG THE FOUR VALUES OVER 110 RUNS (FRIEDMAN $p < 0.001$), AND W/T/L AGAINST THE DEFAULT $\kappa = 2$ WITH SIGN-TEST p.

	Mean fitness				vs. $\kappa = 2$	
κ	Clinical	Gene	Tabular	Rank	W/T/L	p
$\kappa = 1$	0.1836	0.0793	0.1422	**2.20**	54/6/50	0.769
$\kappa = 2$ (default)	0.1832	0.0786	0.1494	2.21	–	–
$\kappa = 3$	0.1855	0.0910	0.1490	2.80	38/1/71	**0.002**
$\kappa = 4$	0.1857	0.0882	0.1525	2.80	35/5/70	**<0.001**

TABLE XIII
MEDICAL TABULAR DATASETS WITH AN SVM AND WITH 5-NN AS THE CLASSIFIER (50 PAIRED RUNS EACH, SAME SPLITS AND SEEDS): W/T/L FOR THE FIRST-NAMED METHOD ON FITNESS AND UNADJUSTED TWO-SIDED SIGN-TEST p. BOLD: $p < 0.05$.

	SVM		5-NN	
Comparison	W/T/L	p	W/T/L	p
RL-DOHO vs. DO	50/0/0	**<0.001**	49/0/1	**<0.001**
RL-DOHO vs. HO	32/0/18	0.065	30/0/20	0.203
RL-DOHO vs. DOHO	40/2/8	**<0.001**	35/0/15	**0.007**
GA vs. RL-DOHO	43/2/5	**<0.001**	45/1/4	**<0.001**
RL-DOHO vs. BPSO	20/3/27	0.382	20/0/30	0.203
RL-DOHO vs. BGWO	21/2/27	0.471	20/0/30	0.203
RL-DOHO vs. LASSO	41/0/9	**<0.001**	38/0/12	**<0.001**
HO+SI vs. HO	37/5/8	**<0.001**	40/2/8	**<0.001**
GA+SI vs. GA	16/17/17	1.000	18/11/21	0.749
PERTURB only vs. RL-DOHO	41/1/8	**<0.001**	39/3/8	**<0.001**

did help DO: with a 10× or 100× step it beat the published DO in 83 and 84 of 100 runs and evaluated far more distinct subsets (193–382 per tabular run with the 100× step, against 22–26). The standard sigmoid transfer function helped less (60 against 40, $p = 0.057$) and, on gene data, selected hundreds to thousands of genes. RL-DOHO nevertheless stayed ahead of every variant: 75 against 25 runs for the 10× step, 85 against 15 for the sigmoid, and 61 against 39 for the 100× step (Holm $p = 0.035$). The last was level on the tabular data (24 against 26) and decided by the gene data (37 against 13). When both used the 100× step, the layer still improved DO in 73 runs against 20 ($p < 0.001$), on tabular and gene data alike. The published step size is therefore part of why DO is weak, but it is not the reason RL-DOHO beats it, and RL-DOHO itself was unchanged by the larger step (50 against 49). No test-accuracy difference in these comparisons was significant. Supplementary Section S6 gives the details.

B. The interventions help optimizers that lack local moves

The add-on experiment separates the interventions from DO. Attached to HO, they improved fitness in 71 of 100 decided runs; attached to the GA, they changed nothing. The difference is not that the GA never stalls: GA+SI made 47–48 intervention decisions per gene or tabular run, against 12–13 for HO+SI. It lies in what the base optimizer does after a stall. A GA generation recombines and mutates good subsets directly in the binary space, and within GA+SI's interventions the GA arm itself produced 57% of the fitness reduction, so the extra moves largely duplicated what the GA already does. An HO iteration costs about $3N$ evaluations and moves continuous positions relative to a dominant agent; it does not search next to the best subset, which is what PERTURB adds at a cost of $N/2$. The interventions therefore help when they supply a kind of move the base optimizer lacks, rather than wrapper search in general.

Among the interventions, PERTURB, a bit-flip local search around the best subset, is the most useful on low-dimensional data. It produced 69% of the fitness reduction during HO+SI's interventions (42% in GA+SI), and PERTURB alone beat the full arm set on the tabular data with both classifiers (39 against 8 runs with 5-NN, 41 against 8 with an SVM) and was ahead on the clinical data too, although not significantly (6 against 2). On gene data, where a few bit flips change little among thousands of features, the full arm set was ahead (30 against 19, not significant), and in RL-DOHO's main gene runs DO exploration steps produced more of the improvement than PERTURB. The κ sweep points the same way: intervening after one or two stalled iterations was better than waiting three or four. A practical reading is that a stalled population-based

search should be followed quickly by a short local search around its best subset; on our tabular datasets (19–44 features) that local search alone was the better choice overall, although not on every dataset (for example Hepatitis with the SVM, 5 against 4), and on gene data it was not. The fitness advantage of PERTURB alone did not show up in test accuracy (19 against 26 runs with 5-NN, 21 against 22 with an SVM), in line with the other comparisons.

C. Learned move selection does not outperform random choice

The controller result now holds for three controllers (UCB1, a stateful Q-learner and random selection), three reward designs, three data types and two budgets, so it is unlikely to be an artefact of a single setting. Two explanations seemed plausible. The arms might be complementary rather than ranked, so that a random choice loses little. Or the decisions might be too few. At the standard budget the bandit made 9 decisions per clinical run, four of them initial pulls, and about 30 per gene or tabular run; the Q-learner, which has six states to fill, had 10 to 35. That leaves little room for a learned preference to pay off within a run. The Q-learner was, if anything, slightly worse than the bandit, which is consistent with it spending about one in six of its few decisions on random exploration. We therefore describe RL-DOHO's controller as an adaptive selector that costs almost nothing and removes the need to fix the move in advance. On our data it delivered no gain over random selection, and the method's benefits come from the interventions and the initialization.

We tested the second explanation directly (H10a). At three times the budget, with about 80–115 decisions per run, a bandit started from the statistics of two earlier runs on the same training data did no better than random selection (46 against 53 runs, $p = 0.55$). It was, if anything, slightly behind the cold-started bandit (39 against 61, unadjusted $p = 0.035$, Holm $p = 0.070$), while costing about 70% more distinct evaluations for its warm-up runs. The cold-started bandit was again level with random selection (51 against 46).

A diagnostic recorded in the same runs shows why more decisions did not help, and it also rules out the first explanation in its simple form. The moves did differ: under random selection their rewards differed on every dataset (Kruskal–Wallis test [53], $p < 0.001$), and the bandit's choice frequencies ranked the moves exactly as their mean rewards did (Spearman rank correlation [54] 1.0 on all ten datasets). The bandit therefore learned what it was rewarded for. The reward, however, did not rank the moves by what matters. DO iterations earned the highest mean reward (0.10–0.13), largely through the population term of the reward, yet improved the best subset in at most 11% of uses, whereas PERTURB and HO did so in up to 28% and 22%. The reward gaps were also small next to UCB1's exploration bonus, so the choices stayed close to uniform (DO 32%, PERTURB 25%, HO 22%, RESTART 20%). Under these two conditions a learned choice cannot do much better than a random one. We then tested both fixes together (H11, fixed after the H10a diagnostics): a reward that credits only improvement of the best subset, with the default exploration constant and with $c = 0.2$. With the new reward the bandit's preferences moved the right way. Its choice frequencies now tracked how often each move improved the best subset (mean Spearman correlation 0.73, and 0.84 with $c = 0.2$, against 0.08 with the original reward), and PERTURB became its most used move (28%, and 36% with $c = 0.2$). Yet neither variant beat random selection (50 against 48 runs, $p = 0.92$; 55 against 42, Holm $p = 0.45$), and the share of decisions that improved the best subset hardly changed (7.1–7.6%, against 7.4% for random choice). The new reward was ahead of the original one in 57 runs and behind in 37 (unadjusted $p = 0.050$, Holm $p = 0.15$), a trend at most. The reason is the size of the signal. Even the best move improved the best subset in at most 28% of its uses, and usually far less, so favouring it changes the expected outcome of a decision only slightly over about 100 decisions. The bandit can learn which move is better; on this problem, being better is not worth much. The other pre-specified extensions, a move pool with removal and swap moves and a contextual bandit (LinUCB [50]), are implemented (Supplementary Section S3) but were not run.

D. RL-DOHO matches HO and trails the GA on fitness

RL-DOHO and HO reached statistically indistinguishable fitness (58 against 49 over 110 runs), although RL-DOHO had the better family rank and was rarely the worst of the four. On the repeated clinical splits HO had the better family rank. The GA reached lower fitness than RL-DOHO more often on every data type (85 against 22 overall). Its uniform crossover recombines good subsets directly in the binary space, and with a mutation rate of $1/D$ it stays close to sparse subsets once it starts there; neither property depends on a continuous-to-binary mapping. On test accuracy the GA and RL-DOHO did not differ (48 against 48).

A natural objection is that the GA could simply be used instead. On fitness alone, the GA is the better choice on these datasets, and we do not claim otherwise. Three results still matter for practice. First, the GA's fitness advantage did not reach held-out data: no test-accuracy difference survived correction. Second, the GA was more expensive. It evaluated 59–66% more distinct subsets than RL-DOHO per run (132 against 83 on the clinical splits, 633 against 381 on the tabular data), which is what counts when each evaluation means training a model. Third, the GA does not need the layer, and the add-on experiment shows why. GA+SI is a memetic GA in the usual sense, a GA with a local search triggered when it stalls, and it was no better than the plain GA (45 against 49). The GA's crossover and mutation already supply the local moves the layer adds. The layer pays off on optimizers that lack such moves, such as DO and HO, and that is the paper's main claim. It is not a claim that RL-DOHO is the best feature selector.

E. Filter fitness on gene data is optimistic

The filters' standing depended on the data type. On gene data LASSO reached lower fitness than every wrapper on every dataset, and mRMR did so on ALL/AML and Lung. On the

tabular and clinical data, with 14–44 features, the wrappers led and LASSO, mRMR and ReliefF trailed RL-DOHO. Part of the filters' fitness advantage on gene data is optimistic, however. Their subset size was chosen by minimizing the same cross-validated score that is reported as fitness, the selection bias described by Ambroise and McLachlan [44]. Glioma shows the effect most clearly: LASSO had the best fitness and the worst test accuracy. On held-out data the differences shrink: in the pooled tests over all 110 runs no test-accuracy difference survived correction, and LASSO's lead over RL-DOHO on macro-F1 was the only test-score difference that did.

F. The clinical data do not separate the methods

On the rheumatic data all methods except the published DO reached test accuracy within about 1.1 points of each other on the original split and, apart from BGWO, within 0.4 points on each new split, and the best subsets were found by many methods. The dataset is too easy to separate the better search methods, which is also why the intervention layer made no difference to HO or the GA there. The two variables that RL-DOHO, the GA, BPSO and BGWO mostly dropped, anti-dsDNA and anti-Sm, are the two with the most missing values (39% and 43%). Their low value in this dataset may therefore reflect missingness and mode imputation rather than clinical irrelevance. Where fitness evaluations are expensive, as with the boosted models used here, RL-DOHO's smaller number of distinct subsets is a practical advantage at equal test accuracy.

G. Limitations

Several limitations qualify these results. The clinical findings come from a single cohort, now evaluated on three train/test splits with five seeds each, so they describe this cohort rather than rheumatic diagnosis in general, and validation on an external cohort is still needed. The tabular datasets, the add-on experiment, the Q-learning controller, the κ sweep and the SVM runs were designed after the main results were known; their hypotheses were fixed before they were run, but the choice of what to test was informed by the earlier findings. The evaluation budgets (252 and 1,020 fitness evaluations) are modest, and only the controllers were examined at a larger budget. Test sets of 15–171 samples on the gene and tabular data make test scores coarse, which is why we give more weight to fitness and to the pooled tests. The tabular benchmark was repeated with an SVM, but the gene-expression and clinical data were each evaluated with a single classifier configuration (5-NN, and XGBoost with gradient boosting on a 4,000-patient subsample), and no classifier was tuned. RL-DOHO's settings other than κ, including ε and the UCB constant, and the Q-learning settings were fixed rather than tuned, which keeps the comparison even but leaves their sensitivity open. Our DO and HO follow the published equations, but the original papers leave some details open (such as the random-group size in HO), so other implementations may behave differently. The cost figures come from one shared cloud runtime. We did not reimplement recent binary HO variants such as Q2HO-MFTV [21], so the comparison with learned strategy selection inside HO rests on our own Q-learning ablation. Two of the tabular datasets (Backache and Hepatitis) are close to chance for 5-NN; Table V shows which conclusions depend on them. Finally, the filter methods chose their subset size on the same cross-validated score that is reported as fitness, which favours them on that measure, and the pooled sign tests treat runs on the same dataset as independent and combine datasets of different size and difficulty. They are best read together with the per-dataset results and the dataset-level ranks, on which RL-DOHO is separated only from DO and the full feature set.

VII. Conclusion

We presented a stagnation-triggered intervention layer for wrapper feature selection and its instantiation, RL-DOHO. RL-DOHO runs the published DO while the best subset improves and, after $\kappa = 2$ stalled iterations, lets a UCB1 bandit choose among a bit-flip search, a partial restart, a DO iteration and an HO iteration. Across eleven datasets and ten seeds, with equal evaluation budgets and every hypothesis fixed in advance, the main findings are:

- **Against DO:** RL-DOHO reached lower fitness in 103 of 110 paired runs (4 losses). Its lead held against DO with $10\times$ and $100\times$ larger Lévy steps (75 against 25 and 61 against 39 runs) and with a sigmoid transfer function (85 against 15). The layer also improved a DO that used the larger step (73 against 20).
- **Against the other baselines:** it beat the fixed DO-then-HO hybrid (81 against 28), binary PSO (70 against 38), ReliefF (84 against 26) and mRMR (68 against 40), although the mRMR result did not hold without the two weakest tabular datasets (48 against 40, $p = 0.46$). It was not significantly different from HO (58 against 49), binary GWO (67 against 43, Holm $p = 0.084$) or LASSO (49 against 61). It had the best average rank in the DO–HO family (1.77, against 1.89 for HO) and was the family's worst member in only 2 of 110 runs. The GA reached lower fitness in 85 runs against 22.
- **The layer on other optimizers:** it improved HO in 71 runs against 29 ($p < 0.001$; 37 against 8 with an SVM) and left the GA unchanged (45 against 49).
- **The controller:** UCB1 and random choice were level (54 against 51), as were Q-learning and random choice at three times the budget (46 against 56). So were a warm-started UCB1 (46 against 53) and a UCB1 rewarded only for improving the best subset (50 against 48). With that reward, the bandit's choices tracked how often each move improved the best subset (Spearman 0.73–0.84, against 0.08 with the original reward), but no move did so in more than 28% of its uses.
- **Trigger and moves:** intervening after one or two stalled iterations beat waiting three or four (Friedman $p < 0.001$). On the tabular data the bit-flip search alone beat the full set of moves (39 against 8 runs with 5-NN, 41 against 8 with an SVM), but not on gene data (19 against 30).
- **Robustness and cost:** replacing 5-NN with an SVM left the ranking of the methods almost unchanged (Spearman

0.99). Over three clinical train/test splits, RL-DOHO differed significantly only from DO and the full feature set. On the clinical and tabular data it evaluated 35–40% fewer distinct subsets than HO or the GA (83 against 131 and 132 per clinical run).

- **Held-out data:** no pooled test-accuracy difference survived correction.

Future work should test the layer on more clinical cohorts with external validation, test learned controllers on move pools whose members differ more in success rate, and examine binary encodings under which the published DO can move more freely.

Data and Code Availability

The rheumatic dataset is openly available (CC BY 4.0) from the source cited in [7]. The microarray datasets are from the scikit-feature repository [28], GSE93272 is available from the NCBI Gene Expression Omnibus, and the five tabular datasets are from PMLB [46]. The notebooks, per-run results, figures and analysis scripts used in this study are available at https://github.com/adj-04/rl-doho.

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